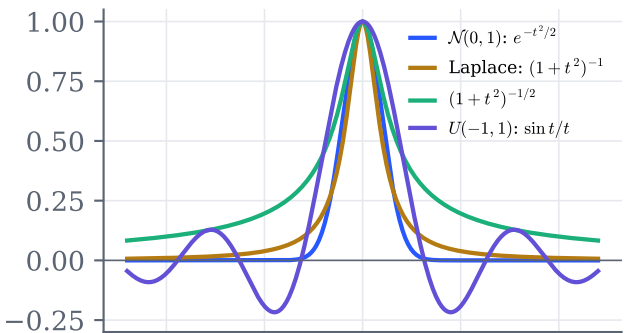
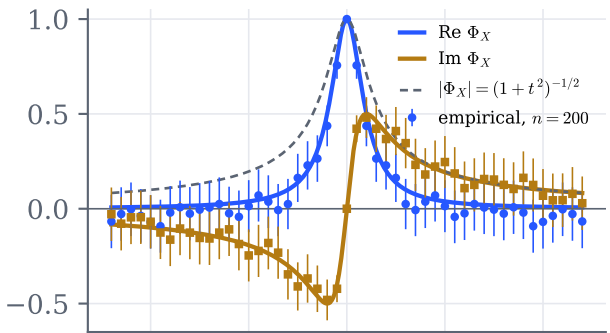

symmetric laws: real Φ_X
1.00
0.75
0.50
0.25
0.00
−0.25
−10
−5
0
5
10
t
$\mathcal{N}(0,1)$: $e^{-t^2/2}$
Laplace: $(1+t^2)^{-1}$
$(1+t^2)^{-1/2}$
$U(-1,1)$: $\sin t/t$


$X \sim \mathrm{Exp}(1)$: $\Phi_X(t) = 1/(1-it)$
1.0
0.5
0.0
−0.5
−10
−5
0
5
10
t
Re Φ_X
Im Φ_X
$|\Phi_X| = (1+t^2)^{-1/2}$
empirical, $n = 200$